

FACE DETECTION SYSTEM USING OPENCV AND HAAR CASCADE CLASSIFIER

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This face detection system operates by initializing a pre-trained Haar Cascade classifier, which serves as the architectural foundation for identifying facial features through pattern recognition. When an image or a webcam frame is processed, the system first converts the raw BGR input into grayscale to reduce computational complexity and eliminate unnecessary color noise. The detection engine then employs the `detectMultiScale` method, which scans the image across multiple scales—controlled by a 1.1 scale factor—to identify face-like features regardless of their distance from the camera. During this scan, the algorithm enforces a strictness threshold via the `minNeighbors` parameter, ensuring that only regions with at least six confirmed overlapping matches are flagged as valid faces. Once a face is detected, the system calculates the coordinates of the match and overlays a bounding box and label onto the original frame in real-time. Finally, the system manages resources by continuously looping through frames in a live feed or processing a static file, utilizing professional logging to monitor the status of the capture hardware and execution flow until a user-initiated exit command is received.